

Automated Data Profiling & LLM-Powered EDA Report Tools

Literature and Landscape Review — May 2026

Section 1 — Academic Papers

1.1 LIDA: A Tool for Automatic Generation of Grammar-Agnostic Visualizations and Infographics using Large Language Models

Authors: Victor Dibia

Year/Venue: 2023 / ACL 2023 (System Demonstrations)

Problem: Systems for automatic visualization must simultaneously understand data semantics, enumerate relevant goals, and generate specifications — tasks historically handled by fragmented pipelines.

Approach: LIDA decomposes visualization as a multi-stage LLM pipeline comprising four modules: a SUMMARIZER (compact NL data summary), a GOAL EXPLORER (visualization goal enumeration), a VISGENERATOR (code generation + repair), and an INFOGRAPHER (stylized graphics via image-generation models). Grammar-agnosticism is achieved by generating visualizations as code in any target language (Matplotlib, Seaborn, Altair, D3, etc.).^[^1]

Relevance: Direct template for the "chart + textual commentary" layer; SUMMARIZER module mirrors the "Senior Data Scientist narrative" design.

Link: <https://arxiv.org/abs/2303.02927>

Tag: [verified]

1.2 Demonstration of InsightPilot: An LLM-Empowered Automated Data Exploration System

Authors: Pingchuan Ma, Rui Ding, Shuai Wang, Shi Han, Dongmei Zhang

Year/Venue: 2023 / EMNLP 2023 System Demonstrations (also arXiv 2304.00477)

Problem: Data exploration requires deep dataset knowledge and analytical expertise, creating bottlenecks for non-expert analysts.

Approach: InsightPilot selects analysis intents (understanding, summarizing, explaining) and concretizes them as Intentional Queries (IQueries) — an abstraction layer over data operations. An LLM iteratively collaborates with a backend insight engine to produce a coherent exploration sequence. A user study validated insight quality.^{[2][3]}

Relevance: Closest architectural ancestor to the "LLM as Senior Data Scientist" design; intent → query → insight pipeline directly applicable.

Link: <https://arxiv.org/abs/2304.00477>

Tag: [verified]

1.3 QUIS: Question-guided Insights Generation for Automated Exploratory Data Analysis

Authors: Abhijit Manatkar, Ashlesha Akella, Parthivi Gupta, Krishnasuri Narayanam

Year/Venue: 2024 / EMNLP 2024 Industry Track

Problem: Fully automated EDA systems either require human goal-specification or demand prohibitive retraining for every new dataset.

Approach: Two-stage pipeline: QUGen generates questions iteratively (refining from prior iterations to maximize coverage), then ISGen analyzes the data to produce multiple insights per question — no prior training required, enabling zero-shot adaptation to new datasets.^[^4]

Relevance: QUGen mirrors the "generate → profile → narrate" loop in the target project; zero-shot adaptability is critical for connecting to arbitrary databases.

Link: <https://arxiv.org/abs/2410.10270>

Tag: [verified]

1.4 Towards Automated Cross-domain Exploratory Data Analysis through Large Language Models (TiInsight)

Authors: Jun-Peng Zhu et al.

Year/Venue: 2024 / arXiv (PingCAP production system)

Problem: Existing LLM-based EDA systems fail on complex database schemas, unclear user intent, and cross-domain generalization.

Approach: TiInsight proposes Hierarchical Data Context (HDC) to compress schema semantics for LLMs, then pipelines HDC generation → question clarification → TiSQL text-to-SQL → TiChart visualization. TiSQL achieves 86.3% execution accuracy on Spider with GPT-4.[^5]

Relevance: Directly solves the "connects to databases" requirement; HDC is a reusable technique for schema understanding in the target project.

Link: <https://arxiv.org/abs/2412.07214>

Tag: [verified]

1.5 Quality Assessment of Tabular Data using Large Language Models and Code Generation

Authors: Ashlesha Akella, Akshar Kaul, Krishnasuri Narayanam, Sameep Mehta

Year/Venue: 2025 / arXiv:2509.10572

Problem: Rule-based data quality validation is brittle, requiring extensive human intervention and struggling with semantic validity.

Approach: A three-stage framework combines statistical outlier detection, RAG-aided LLM rule generation (with domain knowledge sources and few-shot examples), and LLM-synthesized executable validators. Guardrails ensure accuracy and consistency of generated rules and code.[^6]

Relevance: Blueprint for the "data quality issue detection" layer; RAG-grounded rule generation reduces LLM hallucination on quality checks.

Link: <https://arxiv.org/abs/2509.10572>

Tag: [verified]

1.6 LLM-Based Data Science Agents: A Survey of Capabilities, Challenges, and Future Directions

Authors: Mizanur Rahman, Amran Bhuiyan, Mohammed Saidul Islam, et al.

Year/Venue: 2025 / arXiv:2510.04023

Problem: No comprehensive lifecycle-aligned taxonomy existed for LLM-based data science agents across the full DS workflow.

Approach: Maps 45 systems onto six lifecycle stages (business understanding, EDA/visualization, feature engineering, model building, interpretation, deployment) and annotates five design dimensions (reasoning style, modality, tool orchestration, alignment, trust/safety). Key finding: over 90% of surveyed systems lack explicit trust and safety mechanisms.[^7]

Relevance: Essential gap map; confirms that "EDA + visualization" is the most covered stage while data quality, drift, and deployment are underserved.

Link: <https://arxiv.org/abs/2510.04023>

Tag: [verified]

1.7 A Survey on Large Language Model-based Agents for Statistics and Data Science

Authors: (Multiple, Chinese academy consortium)

Year/Venue: 2024 / arXiv:2412.14222

Problem: LLM-based "data agents" lack a unified framework characterizing their planning, reflection, multi-agent, and UI design patterns.

Approach: Reviews evolution and capabilities of LLM data agents; analyzes planning, reasoning, reflection, multi-agent collaboration, and knowledge integration; provides case studies on real-world deployments including conversational EDA systems.[^8]

Relevance: Broad survey covering multi-agent orchestration patterns directly applicable to the "Senior Data Scientist" agent architecture.

Link: <https://arxiv.org/abs/2412.14222>

Tag: [verified]

1.8 A Data-centric AI Framework for Automating Exploratory Data Analysis and Data Quality Tasks

Authors: (IBM Research team)

Year/Venue: 2023 / ACM Transactions on Database Systems (TODS, published Nov 2023)

Problem: Data scientists spend the majority of their time on data preparation, with no unified framework spanning EDA and data quality for AI pipelines.

Approach: Proposes four novel algorithms addressing EDA and data quality pain points (identified via user interviews with data scientists); implements a tool that auto-generates Python code embedding these algorithms. User study showed 2× productivity gain and 30–50% time savings.[^9]

Relevance: Validates the productivity case for automated EDA; the code-generation approach (rather than direct LLM narration) is an alternative design path worth benchmarking.

Link: <https://dl.acm.org/doi/10.1145/3603709>

Tag: [verified]

1.9 NL4DV: A Toolkit for Generating Analytic Specifications for Data Visualization from Natural Language Queries

Authors: Arpit Narechania, Arjun Srinivasan, John Stasko

Year/Venue: 2020 / IEEE VIS 2020 (IEEE TVCG)

Problem: Building NL-to-visualization interfaces requires combined NLP and visualization design expertise, creating a high barrier.

Approach: NL4DV takes a tabular dataset and NL query and returns a JSON specification containing detected data attributes, inferred analytic tasks, and Vega-Lite visualization recommendations. Enables rapid prototyping of visualization NLIs without NLP expertise.[^10]

Relevance: Foundational toolkit for the visualization recommendation layer; the LLM extension (arXiv:2408.13391) updates this to GPT-4-based analytic spec generation.

Link: <https://arxiv.org/abs/2008.10723>

Tag: [verified]

1.10 Generating Analytic Specifications for Data Visualization from Natural Language Queries using LLMs (NL4DV-LLM)

Authors: Subham Sah, Rishab Mitra, Arpit Narechania, Alex Endert, John Stasko, Wenwen Dou

Year/Venue: 2024 / NLVIZ Workshop @ IEEE VIS 2024

Problem: LLM-based NL-to-visualization approaches are opaque "black boxes" lacking explainability and debuggability of the translation process.

Approach: Extends NL4DV with a GPT-4-backed prompt that returns a full analytic specification (data attributes mapped to query phrases, visual design justifications), supports conversational interaction and ambiguity detection.[^11]

Relevance: Directly solves the "why this chart?" explainability problem for the visualization recommendation engine.

Link: <https://arxiv.org/abs/2408.13391>

Tag: [verified]

1.11 Data-centric Artificial Intelligence: A Survey

Authors: Daochen Zha, Zaid Pervaiz Bhat, Kwei-Herng Lai, Fan Yang, et al.

Year/Venue: 2023 / arXiv:2303.10158 (revised Jun 2023)

Problem: The AI community lacks a holistic framework covering the full data lifecycle — training data development, inference data development, and data maintenance.

Approach: Surveys methods across three data-centric goals; organizes literature from automation and collaboration perspectives; tabulates benchmarks for data collection, labeling, quality, and augmentation tasks. [^12]

Relevance: Frames "automated data quality assessment" within the broader data-centric AI paradigm; provides vocabulary and taxonomy for positioning the project.

Link: <https://arxiv.org/abs/2303.10158>

Tag: [verified]

1.12 Can LLMs Narrate Tabular Data? An Evaluation Framework for Natural Language Representations

Authors: Jyotika Singh, Weiyi Sun, Amit Agarwal, et al. (Oracle AI)

Year/Venue: 2025 / arXiv:2510.23854

Problem: LLM-generated natural language representations of SQL query results suffer from information loss and errors that are largely unmeasured.

Approach: Introduces Combo-Eval, a multi-method evaluation framework that judges LLM-generated NL representations of tabular results, combining existing evaluation approaches to optimize fidelity assessment.
[^13]

Relevance: Directly addresses the "hallucination on statistics" risk for the narrative generation layer; Combo-Eval could be adapted as a quality gate.

Link: <https://arxiv.org/abs/2510.23854>

Tag: [verified]

1.13 Why Do Open-Source LLMs Struggle with Data Analysis? A Systematic Empirical Study

Authors: Yuqi Zhu, Yi Zhong, Jintian Zhang, Ziheng Zhang, et al.

Year/Venue: 2025 / arXiv:2506.19794

Problem: Open-source LLMs underperform on data analysis tasks despite strong general benchmarks; root causes were unclear.

Approach: Curates a diverse set of realistic data analysis scenarios; evaluates models across three dimensions — data understanding, code generation, and strategic planning. Key finding: strategic planning quality is the primary performance determinant; interaction design significantly influences outcomes.
[^14]

Relevance: Critical for model selection decisions; open-source vs. API trade-offs for the "Senior Data Scientist" LLM backend.

Link: <https://arxiv.org/abs/2506.19794>

Tag: [verified]

1.14 InReAcTable: LLM-Powered Interactive Visual Data Story Construction from Tabular Data

Authors: Gerile Aodeng, Guozheng Li, Yunshan Feng, et al. (Beijing Institute of Technology)

Year/Venue: 2025 / arXiv:2508.18174

Problem: Constructing visual data stories from tabular data requires sifting through many discrete insights and integrating them into coherent narratives — a complex human task.

Approach: Proposes an LLM-powered interactive system that extracts data insights, identifies relationships between them, and constructs visual data stories through user-guided interaction, enabling iterative narrative refinement.
[^15]

Relevance: Models the "narrative findings" output layer; interaction design for user-guided story construction is directly relevant to report UX.

Link: <https://arxiv.org/abs/2508.18174>

Tag: [verified]

1.15 Large Language Model-based Data Science Agent: A Survey

Authors: Peiran Wang, Yaoning Yu, Ke Chen, Xianyang Zhan, Haohan Wang

Year/Venue: 2025 / arXiv:2508.02744

Problem: Existing surveys on LLM agents lack a data science-specific dual-perspective framework connecting agent design principles with DS workflows.

Approach: Comprehensive analysis of LLM-based data science agents covering agent roles, execution strategies, knowledge and reflection methods from the agent design perspective, and data preprocessing, model development, evaluation, visualization from the DS workflow perspective.
[^16]

Relevance: Most up-to-date (Aug 2025) review of the target design space; recommended as a secondary survey after arXiv:2510.04023.

Link: <https://arxiv.org/abs/2508.02744>

Tag: [verified]

Section 2 — GitHub Repositories

Data Profiling Libraries

Name	GitHub URL	Stars (observed May 2026)	One-line description	Activity	Strengths vs. project	Gaps vs. project
ydata- profiling	https://github.com/ydataai/ydata-profiling	~13,500 ^[^17]	One-line EDA report generator for pandas DataFrames producing rich HTML with statistics, distributions, and correlations	Active (updated Jan 2026) ^[^18]	Gold standard for descriptive statistics + column profiles; data type inference; correlation matrices	No LLM narrative; no DB connector; no schema drift detection; static output only
Sweetviz	https://github.com/fbdesignpro/sweetviz	~3,100 ^[^19]	Beautiful, high-density HTML EDA reports with emphasis on target variable comparison and train/test dataset diffing	Active (updated Apr 2026) ^[^19]	Strong target-analysis and dataset comparison; self-contained HTML	No LLM integration; no database support; no DQ anomaly detection
DataPrep	https://github.com/sfu-db/dataprep	~2,100 ^[^20]	Fast Dask-based EDA toolkit (10× faster than pandas-based profiling) with clean and connector modules	Moderately active ^[^21]	Speed advantage for large DataFrames; <code>dataprep.clean</code> for type standardization; <code>dataprep.connector</code> for APIs	No LLM narrative; no outlier/anomaly scoring; connector is API-focused, not SQL DB
D-Tale	https://github.com/man-group/dtale	~4,500 ^[^22]	Interactive Flask+React GUI for pandas DataFrames with drill-down charts, correlations, and code export	Active ^[^23]	Rich interactive UI; code export for reproducibility; correlation heatmaps	No LLM; no automated narrative; interactive-only (no batch report generation)
DataProfiler (Capital One)	https://github.com/capitalone/DataProfiler	~1,100 ^[^24]	Enterprise Python profiler with schema extraction, statistics, and PII/sensitive data detection via deep learning	Moderately active ^[^25]	Unique PII/NPI detection; handles mixed-type columns; supports CSV, JSON, Parquet	No LLM narrative; limited visualization; stale (few recent commits) ^[^26]
klib	https://github.com/akanzi/klib	~900 ^[^27]	Lightweight Python library for importing, cleaning, analyzing, and preprocessing data with opinionated visualizations	Stale (last updated 2023) ^[^27]	Clean one-line charts (missing values, correlations, distributions); pandas-native	No LLM; very limited profiling scope; stale development

LLM-Powered Data Agents

Name	GitHub URL	Stars (observed May 2026)	One-line description	Activity	Strengths vs. project	Gaps vs. project	Tag
PandasAI	https://github.com/sinaptik-ai/pandas-ai	~16,000+ [^28]	NL-to-analysis platform letting users chat with DataFrames and databases in natural language	Active (YC W24 startup) [^29]	Multi-LLM support; chart generation; connects to various DB types	No automated columnar profiling; no DQ report; chat-first UX not report-first	[verified]
Vanna AI	https://github.com/vanna-ai/vanna	~23,500[^30]	RAG-based framework for accurate Text-to-SQL with self-learning from query history	Active (updated Feb 2026)[^30]	Best-in-class Text-to-SQL accuracy; connects to 20+ databases; self-learning	Focused on SQL query answering only; no profiling, DQ, or narrative report generation	[verified]
LIDA	https://github.com/microsoft/lida	~4,500[^31]	Grammar-agnostic visualization generator from data using LLMs (summarize → goal → code → infographic)	Moderately active[^32]	SUMMARIZER directly produces compact data description; grammar-agnostic chart code	No columnar DQ checks; no database connector; limited to single-dataset exploration	[verified]
Sketch	https://github.com/approximatelabs/sketch	~2,600[^33]	AI code-writing assistant that understands pandas DataFrame context for NL querying and code suggestion	Stale (last major update 2023)[^33]	Elegant .sketch.ask and .sketch.howto interface; lightweight; shows generated code	No report generation; no visualization; no DQ pipeline; unmaintained	[verified]
Jupyter AI	https://github.com/jupyterlab/jupyter-ai	~4,000+[^34]	Official JupyterLab extension integrating frontier LLMs via %ai magic and a native chat panel	Active (v3 in development) [^35]	Vendor-neutral; supports local models (Ollama, GPT4All); notebook-native	Not a profiling or reporting tool; no automated EDA; requires notebook environment	[verified]

Auto-Visualization / Narrative Reporting

Name	GitHub URL	Stars (observed May 2026)	One-line description	Activity	Strengths vs. project	Gaps vs. project	Tag
AutoViz	https://github.com/AutoViML/AutoViz	~1,600[^36]	One-line automatic visualization for any dataset, any size; now includes FixDQ() for data quality assessment	Active[^37]	Broad chart types; FixDQ() gives DQ score; Bokeh/Holoviews interactive mode	No LLM narrative; limited DQ depth; no database connector	[verified]

Name	GitHub URL	Stars (observed May 2026)	One-line description	Activity	Strengths vs. project	Gaps vs. project	Tag
NL4DV	https://github.com/arpit-omprakash/nl4dv (Python package at PyPI)	~200[^38]	Python toolkit mapping NL queries to Vega-Lite visualization specifications with analytic task tagging	Maintained[^38]	Explainable analytic spec output; maps NL entities to query phrases; open-source and extensible	Pre-LLM rule-based NLP (though LLM extension exists); no data profiling	[verified]

Data Quality / Anomaly Detection Libraries

Name	GitHub URL	Stars (observed May 2026)	One-line description	Activity	Strengths vs. project	Gaps vs. project	Tag
Great Expectations	https://github.com/great-expectations/great_expectations	~12,000 ^[39] [40]	Declarative data quality testing framework with Expectations (unit tests for data) and Data Docs HTML reports	Active[^41]	Industry standard; covers missingness, schema, uniqueness, distributions, referential integrity; CI/CD integration	No LLM narrative; schema-drift only via manual versioning; steep learning curve	[verified]
DataKitchen TestGen	https://github.com/DataKitchen/dataops-testgen	~1,500[^42]	Open-source (Apache 2.0) data quality tool that auto-generates test suites via profiling, then monitors in production	Active (announced 2025)[^43]	Auto-generates ~60 DQ tests from profiling; anomaly monitoring; quality scorecards; no per-table fees	No LLM narrative generation; limited visualization; no database-agnostic connector layer	[verified]

Full-Stack EDA Report Generators

Name	GitHub URL	Stars (observed May 2026)	One-line description	Activity	Strengths vs. project	Gaps vs. project	Tag
TiInsight (PingCAP)	https://github.com/pingcap/tiinsight	[uncertain — not publicly listed as of survey date]	End-to-end SQL-based EDA system with HDC, TiSQL, TiChart in production at PingCAP	Active (production) [^5]	Cross-domain generalization; SQL-native; full pipeline from schema to chart	Not open-sourced as a reusable library; TiDB-centric; no standalone DQ report layer	[uncertain]
AutoEDA (Devang-C)	https://github.com/Devang-C/AutoEDA	~200[^44]	Streamlit web app for interactive EDA: dataset overview, distribution charts, correlation heatmaps	Stale (2023) [^44]	Quick web-based EDA; covers missing data and correlation visualization	No LLM; no DQ scoring; no database connector; discontinued	[verified]

Section 3 — Commercial / Closed-Source Tools

Vendor	Description	Overlapping Capability	Pricing Tier
Julius AI	AI-powered data analyst; upload CSV/Excel/Google Sheets, ask NL questions, get charts and reports ^{[45][46]}	NL querying, chart generation, narrative summaries, multi-format file support	Free (15 req/mo); Basic \$20/mo (250 queries); Essential \$45/mo (unlimited) ^[^47]
Hex	Collaborative SQL+Python notebook workspace with "Hex Magic" AI suite: NL query generation, Notebook Agent, code debugging, semantic models ^{[48][49]}	SQL/Python EDA, NL-to-code, interactive dashboards, data apps	\$25/user/month (Starter); enterprise custom; raised \$70M (May 2025) ^[^48]
Tableau Pulse	Proactive AI metrics layer delivering personalized automated insights, NL Q&A, and root-cause analysis via Slack/email/mobile ^{[50][51]} ^[^52]	Auto-insight generation, NL querying, KPI monitoring, metric bootstrapping	Part of Tableau subscription (\$75/user/mo Tableau Standard) ^[^53]
Julius AI <i>(duplicate removed — see above)</i>	—	—	—
Mode Analytics	Collaborative SQL/Python/R analytics platform with AI Assist for SQL generation, query explanation, and chart editing via NL ^{[54][55]}	Ad hoc SQL + Python EDA, governed datasets, AI SQL assist, shared dashboards	Contact for pricing; acquired by ThoughtSpot ^[^54]
Akkio	No-code ML + Generative BI platform for non-technical users; Chat Explore for conversational data analysis, live dashboards, forecasting ^{[56][57]}	Chat-based data exploration, Generative BI dashboards, no-code ML predictions	Basic \$49/user/mo; Pro \$99/user/mo; Build-On from \$999/mo; Enterprise custom ^[^56]
Athenic AI	Autonomous always-on AI analytics platform connecting to databases/data warehouses; NL querying, AI-suggested follow-up questions, real-time monitoring ^{[58][59]}	NL-to-SQL, multi-source database connectivity, insight discovery, no SQL required	Pricing not publicly listed; free trial available ^[^60]
DataKitchen (Enterprise)	Enterprise tier above open-source TestGen, adding collaboration, governance integration, and flat-rate pricing for unlimited tables ^[^43]	Full-stack DQ profiling + test generation + pipeline observability + quality dashboards	Enterprise custom pricing ^[^43]

Section 4 — Gap Analysis

4.1 Well-Covered Capabilities

The following areas are addressed by multiple mature tools:

- **Statistical profiling of individual columns** (distributions, null rates, cardinality, type inference): ydata-profiling, Sweetviz, DataPrep, D-Tale all do this well for in-memory DataFrames.^[^20]
- **Rule-based data quality validation** (schema checking, uniqueness, range constraints): Great Expectations and DataKitchen TestGen are production-grade.^{[61][43]}
- **Text-to-SQL for database querying**: Vanna AI (23.5k stars) is best-in-class for RAG-based SQL generation.^[^30]
- **Natural language chart generation**: LIDA and PandasAI cover this well for on-the-fly visualization.^{[31][62]}
- **Conversational EDA in notebooks**: Jupyter AI (via %ai magic) and Sketch cover lightweight notebook-native NL analysis.^{[33][34]}

4.2 Weakly Covered or Missing Capabilities

The following capabilities are not integrated in any single open-source tool and are only partially addressed in commercial products:

1. **DB-native automated column-level profiling with LLM narrative in a single pipeline**: No open-source tool combines SQL database connectivity, automated column statistics, and LLM-generated narrative findings in a single deployable report. Existing tools are either DataFrame-only (ydata-profiling) or SQL-query-only (Vanna).^{[20][30]}
2. **Automated schema drift detection with explanatory narrative**: Current drift tools (Great Expectations, DataKitchen) flag changes but produce no LLM-generated explanation of root cause or business impact.^{[42][61]}
3. **Integrated cardinality problem and type inconsistency detection + narrative**: ydata-profiling flags these statistically, but no tool combines detection with LLM-reasoned recommendations for resolution.

4. **Trust & safety mechanisms in generated-code pipelines:** The comprehensive survey (arXiv:2510.04023) found >90% of data science agents lack explicit trust and safety mechanisms — a gap shared by almost every tool reviewed.^[^7]
5. **Report portability and versioning:** No open-source EDA report generator produces versioned, diffable reports suitable for CI/CD data quality gates.

4.3 Strongest Differentiators to Pursue

The combination of the following three capabilities is entirely absent from any single open-source tool and only partially present in commercial tools:

1. **DB-first design with automatic schema understanding** (like TiInsight's HDC, but open and database-agnostic) — covering PostgreSQL, MySQL, SQLite, CSV simultaneously.
2. **Integrated DQ scoring across six dimensions** (missing values, outliers, type inconsistencies, duplicates, cardinality, schema drift) with per-column severity rankings, not just flag/no-flag.
3. **LLM "Senior Data Scientist" narrative layer** that produces analyst-quality prose — including prioritized recommendations, not just observations — grounded in the computed statistics (not hallucinated).

This combination would position the project clearly above ydata-profiling (no LLM, no DB), InsightPilot (no DQ layer), and Julius/Akkio (no open-source, no structured DQ report format).

4.4 Known Technical Risks Not Solved by Existing Tools

LLM hallucination on computed statistics: The closest mitigation (from arXiv:2509.10572 and the Oracle NLR evaluation paper) is to ground LLM prompts with verified statistical outputs and use multi-agent critic patterns. However, no production open-source tool implements this end-to-end. LLMs routinely fabricate statistics when context windows overflow column data.^{[63][6][^13]}

Scale limits on large tables: Most DataFrame-based profilers (ydata-profiling, Sweetviz) load entire tables into memory. DataPrep's Dask backend is the only exception. For tables with hundreds of millions of rows, in-memory profiling is infeasible; sampling strategies introduce statistical bias that the LLM may misinterpret.^[^20]

Code execution security in LLM-generated analysis pipelines: NVIDIA's AI red team documented Remote Code Execution (RCE) vulnerabilities in AI analytics pipelines where LLM-generated Python runs without sandboxing. No open-source EDA agent implements mandatory containerized execution. This is an unresolved production risk for any tool connecting to live databases.^[^64]

Prompt injection via column data: When column names or cell values are included verbatim in LLM prompts (as most tools do), adversarially crafted data values can override system prompts. No reviewed tool implements defense against this attack vector.^[^65]

Inconsistent NL quality across dataset domains: The empirical study (arXiv:2506.19794) shows strategic planning quality — not raw LLM capability — determines output quality. A generic "Senior Data Scientist" persona prompt will degrade significantly on domain-specific datasets (financial, biomedical, etc.) without domain-adapted prompting.^[^14]

Section 5 — Suggested Reading Order

1. arXiv:2510.04023 — *LLM-Based Data Science Agents: A Survey of Capabilities, Challenges, and Future Directions* (Rahman et al., 2025)^[^7]
Why first: Maps 45 systems onto the full DS lifecycle with explicit gap analysis; gives the clearest positioning map for where the target project sits and what over 90% of existing agents are missing.
2. arXiv:2303.02927 — *LIDA* (Dibia, ACL 2023)^[^1]
Why second: The most influential open-source blueprint for the visualization + narrative layer; understanding LIDA's 4-module architecture is prerequisite for designing the chart generation component.
3. arXiv:2304.00477 — *InsightPilot* (Ma et al., EMNLP 2023)^[^2]
Why third: The closest architectural ancestor for the "LLM as analyst" intent → IQuery → insight pipeline; reading this alongside LIDA gives the full EDA agent design space.
4. arXiv:2509.10572 — *Quality Assessment of Tabular Data using LLMs and Code Generation* (Akella et al., 2025)^[^6]
Why fourth: Most directly relevant paper for the data quality detection layer; the RAG + critic-agent pattern is the best-available mitigation against hallucination on quality rules.
5. arXiv:2412.07214 — *TiInsight* (Zhu et al., 2024)^[^5]
Why fifth: Solves the hardest engineering problem (cross-domain database connectivity + schema

comprehension for LLMs) with production-validated results; HDC is an immediately implementable technique for the DB connector module.

References

1. [LIDA: A Tool for Automatic Generation of Grammar-Agnostic ... - arXiv](#) - We present LIDA, a novel tool for generating grammar-agnostic visualizations and infographics. LIDA ...
2. [Demonstration of InsightPilot: An LLM-Empowered Automated Data ...](#) - We introduce InsightPilot, an LLM (Large Language Model)-based, automated data exploration system de...
3. [InsightPilot: An LLM-Empowered Automated Data Exploration System](#) - We demonstrate the effectiveness of InsightPilot in a user study and a case study, showing how it ca...
4. [QUIS: Question-guided Insights Generation for Automated ... - arXiv](#) - We introduce QUIS, a fully automated EDA system that operates in two stages: insight generation (ISG...
5. [Towards Automated Cross-domain Exploratory Data Analysis through Large Language Models](#) - Exploratory data analysis (EDA), coupled with SQL, is essential for data analysts involved in data e...
6. [Quality Assessment of Tabular Data using Large Language Models ...](#) - We present a three-stage framework that combines statistical inliner detection with LLM-driven rule ...
7. [LLM-Based Data Science Agents: A Survey of Capabilities, Challenges, and Future Directions](#) - Recent advances in large language models (LLMs) have enabled a new class of AI agents that automate ...
8. [A Survey on Large Language Model-based Agents for Statistics and Data Science](#) - In recent years, data science agents powered by Large Language Models (LLMs), known as "data agents,...
9. [A Data-centric AI Framework for Automating Exploratory Data Analysis and Data Quality Tasks](#) - Democratisation of machine learning (ML) has been an important theme in the research community for t...
10. [NL4DV: A Toolkit for Generating Analytic Specifications for Data ...](#) - We present NL4DV, a toolkit for natural language-driven data visualization. NL4DV is a Python packag...
11. [Generating Analytic Specifications for Data Visualization from ... - arXiv](#) - The prompt is open-source and integrated into NL4DV, a popular Python-based natural language toolkit...
12. [\[2303.10158\] Data-centric Artificial Intelligence: A Survey - arXiv](#) - In this survey, we discuss the necessity of data-centric AI, followed by a holistic view of three ge...
13. [Can LLMs Narrate Tabular Data? An Evaluation Framework ... - arXiv](#) - Currently, NLR generation is typically handled by large language models (LLMs), but information loss...
14. [Why Do Open-Source LLMs Struggle with Data Analysis? A ... - arXiv](#) - Large Language Models (LLMs) hold promise in automating data analysis tasks, yet open-source models ...
15. [InReAcTable: LLM-Powered Interactive Visual Data Story ... - arXiv](#) - The tabular data consists of many data insights that are valuable patterns and highlight key statist...
16. [Large Language Model-based Data Science Agent: A Survey](#) - The rapid advancement of Large Language Models (LLMs) has driven novel applications across diverse d...
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